Project Portfolio Budgeting as a Markov Decision Process: Environment Design, Baseline Benchmarks, and Optimality Gap Analysis

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Abstract

Allocating a shared capital reserve across a portfolio of simultaneously executing contracted projects is a high-frequency, high-stakes decision for engineering contractors. Classical approaches either assume full foresight or collapse the portfolio to a single project, leaving practitioners with no principled tool for the sequential, multi-project setting under real-world uncertainty. This paper closes that gap by delivering a complete, verified simulation environment that makes the problem tractable for data-driven methods.

We formalise the problem as a Markov Decision Process in which a portfolio manager allocates budget fractions across N concurrent projects at each discrete period. The state encodes Earned Value Management indicators, S-curve planned progress, milestone payment lookaheads, and a per-project breach-tolerance counter. The transition dynamics reproduce the full FIDIC contract cash-flow cycle: advance payment and recovery, progress-weighted milestone gross payments, retention withholding and release, interest charges on treasury draws, and a two-condition termination gate. The reward signal is the γ-discounted portfolio net cash flow, accumulating to an episode NPV that serves as the primary performance criterion. A full-foresight oracle bound is derived analytically, and the exact stochastic optimisation problem is shown to be intractable at portfolio scale via scenario-tree enumeration, formally motivating the MDP/RL formulation over classical stochastic programming.

The environment is released as a configurable, Gymnasium-compatible package —verified across 51 checks in 10 test sections— with a SQLite audit ledger that records every allocation, cash-flow event, and breach flag, eliminating the black-box problem inherent to learned policies. To establish a performance baseline and motivate the need for learned agents, we run three deterministic greedy heuristics (Reach-Plan, Catchup-T, and Catchup-Next-T) and a manually computed full-information optimum across four benchmark configurations of increasing portfolio complexity. The best-heuristic optimality gap widens systematically with complexity, from 3.40 NPV units on an isolated single-project instance to 72.15 units on a three-project portfolio with overlapping cash flows — a 21-fold increase — and on the most complex configuration every greedy heuristic yields a loss while the full-information optimum remains profitable. These results confirm that standard rule-based allocation is structurally insufficient under shared-budget coordination and establish a quantified, well-defined improvement target for reinforcement learning agents. Training such agents (e.g. via PPO), evaluating them against a formal Linear Programming or

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Dynamic Programming bound, and exploring LLM-augmented allocation policies are identified as the natural next steps, addressed in subsequent work.

Keywords: project portfolio management; budget allocation under performance uncertainty; constrained Markov decision process; reinforcement learning; FIDIC contracts; earned value management; Gymnasium environment; greedy baseline; optimality gap.

1 Introduction

1.1 Motivation and Problem Context

Large infrastructure and engineering projects routinely overrun their budgets and miss their schedules (**??**). When a portfolio of such projects executes simultaneously under a shared capital commitment, the risks compound: a performance shortfall on one project propagates through the shared budget to all others, and the decision of how much to allocate to each project at each period can determine whether the portfolio succeeds or fails.

Despite this, the formal treatment of portfolio-level budget allocation as a sequential decision problem is largely absent from the project portfolio management literature. Research has concentrated on the *selection* stage — which projects to include — while the *operational governance* stage — how to manage a committed portfolio through to completion — has received comparatively little analytical treatment (**???**). The present work fills this gap by delivering a verified simulation environment that makes the problem tractable for data-driven methods and establishing a performance baseline that quantifies the cost of naive allocation.

We consider a portfolio of projects, each governed by a bilateral contract consistent with FIDIC forms (**?**), executing concurrently under a shared capital reserve. A portfolio manager exercises one recurring decision at each discrete planning period: how much budget to allocate to each active project. The formal problem is to find the allocation policy that maximises discounted net cash flow across the portfolio horizon while respecting contract mechanics.

1.2 The Project Management Decision Hierarchy

The present contribution operates at the portfolio governance layer of the three-tier project management hierarchy (**?**). Portfolio selection — which projects to commit to — is treated as an exogenous input, consistent with the two-stage governance model of **?**. At the governance layer, the portfolio manager does not control project execution directly but allocates budget to each project and observes earned value outcomes. Project termination is not a deliberate agent action but an endogenous environment mechanic triggered by one of two independent conditions: a hard deadline, which fires immediately if the episode period reaches the planned finish plus the contractual finish delay cap; or sustained underperformance, in which a tolerance counter decrements each period the project's actual progress lags behind its planned progress by more than the contractual cap, and termination executes when the counter reaches zero.

Project termination differs from episode termination. A project terminates individually and triggers a settlement cash flow; the episode ends only when every project has either completed or been terminated, when the global horizon is reached, or when the portfolio budget is exhausted with no pending inflows.

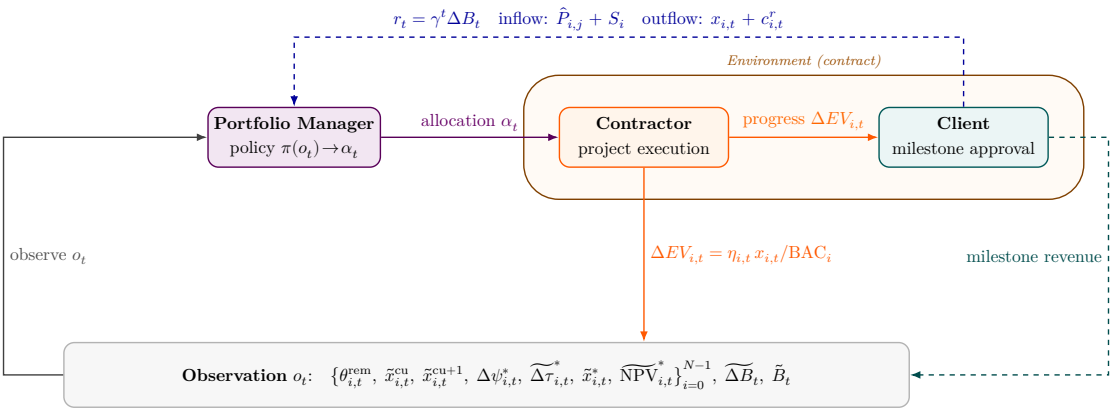


Figure 1: Portfolio governance as a reinforcement learning feedback loop. The portfolio manager commits allocation fractions $\boldsymbol{\alpha}_t$; the contractor converts budget to progress via stochastic efficiency $\eta_{i,t}$; the client certifies milestones and releases net payments $\hat{P}_{i,j}$; the environment returns observation $o_t \in \mathbb{R}^{7N+2}$ and reward r_t. Observation components are defined in full in Table **??**.

1.3 Contract Mechanics and Cash-Flow Structure

The financial interaction between the portfolio manager and each contractor is governed by FIDIC contract forms (**?**). Three mechanisms determine the cash-flow structure.

Milestone-based payment. Payment is contingent on certified progress thresholds. The gross payment at milestone j is a contractually fixed fraction of the contract price. Because payment is contingent on certified progress, the timing of cash inflows depends on the contractor's actual performance and the portfolio manager's allocation decisions. Cash-flow uncertainty in the present model arises from a single source: the stochastic budget-to-progress efficiency $\eta_{i,t}$, which governs how much physical progress a given budget allocation yields in each period. Uncertain productivity induces uncertain milestone timing, which in turn induces uncertain payment arrival — without requiring any separate model of payment delay.

Advance payment and progressive recovery. At project initiation, an advance payment equal to a contractually fixed fraction of the contract price is disbursed to the contractor. It is recovered through percentage deductions from subsequent interim payment certificates, with the commencement threshold and deduction rate specified as project-specific contract parameters (**?**).

Retention. A fraction of each milestone payment is withheld as a performance bond and released in full upon project completion, creating a structural completion incentive embedded in the payment mechanics.

Earned value management as a performance signal. The EVM framework (**?**) provides the diagnostic vocabulary for monitoring contractor performance. The Cost Performance Index and the Earned-Schedule Schedule Performance Index measure cost efficiency and schedule efficiency respectively; the Estimate at Completion uses a composite cost-and-schedule denominator to project total expenditure; and the To-Complete

Performance Index measures the efficiency required on remaining work to finish within budget. These four indicators are tracked internally and logged to the audit record each period, but are not directly exposed in the observation vector; they serve as intermediates from which the catchup-allocation and target-milestone features are derived.

Termination. A distinguishing feature of the proposed environment is that project termination is treated as an endogenous consequence of the manager's own funding decisions, rather than an exogenous shock. In practice, a contractor can lose a project in two qualitatively different ways: gradually, as sustained underfunding allows schedule slippage to persist past a contractually tolerated threshold and through a grace period intended to allow recovery; or abruptly, once the project's finish date overruns its contractually permitted limit regardless of how close it is to catching up. Capturing only one of these failure modes — as prior single-project formulations typically do — understates the manager's true risk exposure. Termination also carries direct financial consequences that must be settled against the portfolio's cash position, since work already paid for and work already completed rarely coincide exactly at the moment a contract is terminated. By embedding both breach conditions, their associated grace mechanism, and the resulting settlement cash flow directly into the transition dynamics, the environment forces any policy to treat the threat of termination as a consequence of its own allocation choices — precisely the incentive structure a real portfolio manager faces.

1.4 The Optimality Gap and the Case for Learned Policies

A central empirical question is whether the problem is hard enough to warrant learned policies at all. To answer this, we run three deterministic greedy heuristics against a manually computed full-information optimum across four benchmark portfolio configurations of increasing complexity.

The **Reach-Plan** heuristic (π^{rp}) allocates exactly enough budget each period to bring every active project to its S-curve planned progress, closing the progress gap entirely. The **Catchup-T** heuristic (π^{cu}) allocates only enough to stay within the contractual progress delay tolerance, minimising expenditure. The **Catchup-Next-T** heuristic ($\pi^{\mathrm{cu}+1}$) is more risk-averse: it pre-emptively allocates enough this period so that the projected delay next period also remains within tolerance. The **full-information optimum** (π^{*}) is computed manually with all stochastic realisations known in advance, serving as an upper-bound reference.

Figure **??** reports cumulative episode NPV for each method across four configurations. On the simplest configuration (single project, fixed efficiency) the greedy baselines come within a small margin of the optimum. As portfolio complexity grows — more projects, staggered starts, interacting cash flows, and milestone sequencing constraints — the gap widens systematically. On the most complex configuration the best greedy baseline trails the optimum by more than 60 NPV units while one baseline degrades to a loss-making outcome. This monotone widening confirms that the allocation problem becomes genuinely difficult at scale and that a well-defined improvement target exists for reinforcement learning agents.

1.5 Contributions and Paper Organisation

The contributions of this paper are threefold.

C1. Environment. A configurable, Gymnasium-compatible MDP environment encoding the full FIDIC contract cash-flow cycle: advance payment and progressive

recovery, progress-weighted milestone payments, retention withholding and release, treasury-draw interest on negative project cash balances, and a two-condition termination gate with a project-specific cure-period counter. All environment parameters are specified through a configuration dictionary whose entries are probability distributions, allowing practitioners in any project-based sector — construction, IT, R&D, or otherwise — to instantiate domain-specific environments by fitting those distributions to historical company data. Gymnasium API compliance is verified across 51 checks in 10 test sections; the full compatibility certificate is reproduced in Appendix **??**. The environment ships with a SQLite audit ledger that records every allocation, cash-flow event, and breach flag at each episode step, making policy behaviour fully interpretable.

C2. MDP Formulation. A formal specification of the portfolio budgeting problem as a Markov Decision Process: state space, action space, transition dynamics, reward signal, and the observation vector presented in closed form, with complete alignment between the mathematical specification and the implemented environment.

C3. Baseline Benchmark and Optimality Gap. Three deterministic greedy heuristics evaluated against a full-information optimum across four benchmark configurations, producing a quantified optimality gap that grows monotonically with portfolio complexity and establishes the performance target for future reinforcement learning agents.

Section 2 reviews the relevant literature across four streams: project management foundations, portfolio management and selection, PPM budgeting and cash-flow modelling, and reinforcement learning for sequential resource allocation. Section **??** develops the MDP formulation formally. Section **??** presents the benchmark evaluation: three deterministic greedy heuristics and a full-information optimum across four portfolio configurations, quantifying the optimality gap that motivates learned agents. Training RL agents on the environment is left as the natural next step, addressed in subsequent work.

2 Literature Review

2.1 Overview and Review Strategy

This section maps the intellectual lineage of the present work across four adjacent knowledge domains: project management (PM) foundations, project portfolio management (PPM) and selection, PPM budgeting and cash-flow modelling, and reinforcement learning (RL) applied to sequential resource-allocation problems. Together these streams define both the theoretical foundation and the gap that motivates the proposed framework. The review follows the PRISMA 2020 reporting guidelines (**?**) to ensure transparency and reproducibility in source identification, screening, and inclusion decisions.

2.1.1 Search and Screening Protocol

Literature was retrieved from Google Scholar, Semantic Scholar, and Scopus using a progressively narrowing search strategy. Queries began at the broadest relevant domain (*project management*) and were successively refined through *project portfolio management*, *portfolio budgeting and cash-flow optimisation*, *earned value management*

under uncertainty, and *reinforcement learning for project scheduling and resource allocation*. Citation-network traversal — backward and forward snowballing from high-citation anchor papers — was used to recover seminal works not surfaced by keyword search.

Records were screened in three sequential automated stages consistent with the PRISMA 2020 flow diagram (Appendix **??**). The screening pipeline, together with its archived corpus and output artefacts, is implemented as a reproducible processing chain whose repository location and concept-level mapping are documented in Appendix **??**.

Stage 1 — Identification and Machine Readability. Candidate PDFs were subjected to automated bibliographic and readability triage. Each document was assessed for text-extraction yield (minimum readable character density across pages), lexical quality (alphabetic-token ratio and mean word length), and sentence-transformer embeddability. Documents failing any criterion were quarantined in a dedicated rejection archive, with a content hash logged to prevent duplicate submissions from re-entering the pipeline.

Stage 2 — Relevance Filtering. Records passing Stage 1 were ranked by a composite relevance score. A semantic similarity component embedded each paper's abstract, introduction, and conclusion using a sentence-transformer model and computed the maximum cosine similarity against twelve thesis-derived query vectors spanning the target knowledge domains. A keyword-scoring component assigned weighted counts of domain-specific terms. The two components were combined with equal weight to yield a combined relevance score in $[0, 1]$. Papers were triaged into *highly relevant* (score ≥ 0.45), *valuable* (score ≥ 0.30), and *irrelevant* (score < 0.30), with the first two cohorts retained for structured extraction.

Stage 3 — Structured Extraction. Each retained paper was subjected to structured knowledge extraction. The abstract, introduction, and conclusion of each PDF were parsed and submitted to a large language model with a fixed schema prompt, producing a per-paper record covering: problem framing, portfolio level, decision type, stochastic cash-flow treatment, contract mechanics, termination-settlement provisions, modelling paradigm, calibration parameters, and thesis-relevance flags. The resulting structured corpus drove an automated synthesis dashboard and table generator, producing the gap-analysis matrix, domain-distribution charts, and the literature map reproduced in Table 1.

The complete pipeline — processing scripts, the structured record corpus, dashboard outputs, and synthesis tables — is archived within the project repository and constitutes a reproducible, machine-auditable literature review corpus; its directory-level organisation is detailed in Appendix **??**. The PRISMA flow diagram quantifying records at each stage is provided in Appendix **??**.

2.2 Project Management: Foundational Concepts

The academic study of project management draws on schedule control (CPM/PERT), earned value management (EVM), and risk modelling as its core quantitative pillars (**??**). EVM introduced the Cost Performance Index and the Schedule Performance Index as real-time diagnostic signals relating budgeted cost of work performed to actual expenditure and planned schedule — providing the performance vocabulary that the present model inherits directly. The present model adopts the Earned Schedule formulation of the Schedule Performance Index (**?**), which continues to degrade past the planned finish date rather than freezing once planned value reaches the budget at

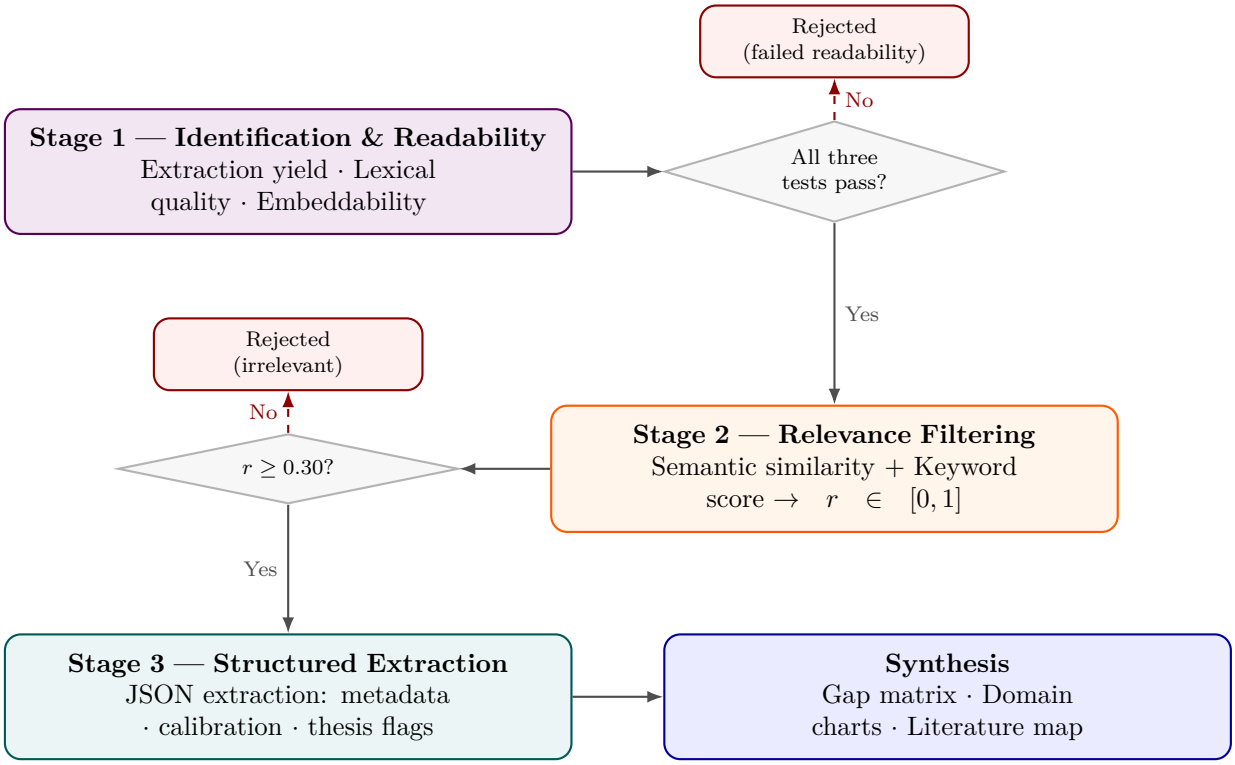


Figure 2: Three-stage PRISMA-aligned screening pipeline, arranged as a serpentine flow. Boxes denote automated processing stages; diamonds denote pass/fail gates; dashed arrows indicate rejection routing. Implementation details and repository locations are provided in Appendix **??**.

completion — a known limitation of the classical ratio in post-planned-finish recovery periods (**??**). The To-Complete Performance Index is included as a forward-looking complement, measuring the cost efficiency required on remaining work to finish within budget. The Estimate at Completion uses a composite cost-and-schedule denominator; **?** established across 155 contracts that this formulation produces the most accurate and stable cost-at-completion forecasts, with reliability increasing beyond the 20% completion mark (**?**). Stochastic extensions of EVM (**??**) and Monte Carlo simulation frameworks (**?**) have modelled performance uncertainty at the project level, establishing that the performance indices follow empirically characterisable distributions over project life and providing the calibration basis for the stochastic efficiency parameter in the present environment.

S-curve analysis, formalised by **?** for construction cash flows, established that budget consumption follows characteristic sigmoid profiles whose shape parameters are project-type- and contract-form-dependent. The regularised incomplete Beta function provides a flexible single-family parameterisation of this profile (**?**) and is adopted directly in the planned expenditure model of the present formulation.

Large-scale empirical evidence on project performance is provided by **?** through the Independent Project Analysis database, and by **??** through infrastructure cost-overrun datasets spanning decades and geographies. These datasets inform both the calibration of uncertainty distributions and the benchmarks against which synthetic environment parameters are validated. Complementary empirical benchmarks are drawn from AACE International and Construction Industry Institute cost-engineering databases (**?**).

Contract governance structures — particularly FIDIC forms (Red, Yellow, and Silver Books) — prescribe the financial interaction between owner and contractor through milestone payment schedules, advance payment and recovery mechanisms, and retention clauses (**?**). Under FIDIC Sub-Clause 14.2, the advance payment is an interest-free mobilisation loan recovered through percentage deductions from interim payment certificates; the commencement threshold and recovery rate are specified as project-specific particular conditions (**?**). The present model parameterises this mechanism with project-specific advance payment and recovery rates, accommodating the full range of particular conditions encountered in practice.

2.3 Project Portfolio Management: Selection and Governance

Project portfolio management formalises the problem of allocating an organisation's finite resources across a competing set of candidate projects to maximise strategic value (**??**). The literature distinguishes two nested decision layers: *portfolio selection* — the binary or mixed-integer optimisation of which projects to include — and *portfolio governance* — the ongoing operational oversight of approved, active projects.

Portfolio selection models have been extensively developed. Markowitz-style mean–variance frameworks adapted from financial portfolio theory (**?**) have been applied to project baskets to trade off expected NPV against variance (**?**). Multi-criteria decision analysis extensions incorporate strategic alignment, risk ratings, and resource interdependencies (**??**). Metaheuristic approaches address large combinatorial instances (**?**). Resource-constrained portfolio selection under capital rationing has been studied by **?**, whose no-borrowing constraint is adopted directly in the present formulation.

The present work explicitly adopts the output of portfolio selection as an exogenous input. The selection problem is treated as solved upstream; the focus is entirely on the operational governance layer that receives a committed portfolio and must manage it through execution. This decomposition is consistent with the two-stage governance model

of **?** and the PMI Standard for Portfolio Management (**?**), which separates strategic selection from operational execution oversight. In doing so, this work builds directly on the selection literature while advancing the frontier into the governance domain that has received substantially less formal analytical treatment.

2.4 PPM Budgeting, Cash-Flow Modelling, and Uncertainty

Within portfolio governance, budget allocation across active projects over time represents a high-frequency, high-stakes operational decision. Despite its practical centrality, this specific problem has received limited formal treatment relative to portfolio selection. Existing work addresses project-level cash-flow modelling (**??**), EVM-based forecasting of individual project cost-at-completion (**??**), and multi-project resource levelling (**??**). These contributions treat budgets as deterministic constraints rather than as dynamic decision variables subject to portfolio-level sequential optimisation.

Cash-flow uncertainty in the present model arises from a single source: the budget-to-progress efficiency, which governs how much physical progress a given budget allocation yields in each period (**??**). Uncertain productivity determines whether and when each milestone threshold is crossed, which in turn determines the timing of the contractually fixed milestone payment. Payment delays between certification and receipt are not modelled as an independent uncertainty source; the performance-driven channel is the structurally primary source of cash-flow timing uncertainty in milestone-payment contracts, and isolating it yields a cleaner identification of the allocation problem without loss of the essential uncertainty framing. Other sources of cash-flow uncertainty — scope changes, external market shocks (**?**) — are treated as exogenous.

Stochastic programming (**?**) provides the canonical OR framework for optimisation under such uncertainty, and its two-stage variant has been applied to project scheduling under resource uncertainty (**?**). However, multi-stage stochastic programs for portfolio-level sequential budget allocation face scenario-tree explosion as the number of projects and periods increases, as formalised in Section **??**. This intractability motivates the MDP formulation of the present work.

The gap in the literature is precise: no established framework models the portfolio-level budget allocation decision as a sequential stochastic optimisation problem in which a central decision-maker allocates funds across multiple active projects whose cash inflow trajectories are uncertain through stochastic budget-to-progress efficiency, subject to contractual milestone certification windows that bound the speed at which progress converts to cash inflows.

The reward function of the present environment is pure discounted net cash flow; no deviation penalty enters the objective. This is a deliberate design choice: the consequences of sustained deviation — termination and loss of all future milestone payments and retained amounts — are sufficient to make deviation undesirable without a separate penalty term, and including one would introduce an arbitrary scalarisation weight with no natural calibration from contract data (**?**). The discount factor encodes not only time preference but the opportunity cost of tied-up capital in a competitive environment where future projects are foregone while current ones remain incomplete (**?**), giving it a concrete calibration anchor from the firm's observed pipeline redeployment rates or prevailing cost-of-capital benchmarks. A consequence is that a well-performing policy may produce allocation trajectories that concentrate expenditure around milestone certification thresholds, mirroring documented contractor financial strategy in milestone-payment contracts (**???**).

2.5 Contractor Underperformance, Cure Periods, and Termination

When a contractor fails to meet schedule or cost performance thresholds, international EPC contracts prescribe a structured escalation sequence before termination may be exercised. Under FIDIC Sub-Clause 15.1, the client must issue a formal notice of default and allow a cure period — typically 28 days — during which the contractor is obliged to demonstrate measurable recovery before the client may proceed to termination (**??**). This cure period is not a discretionary grace; it is a contractual obligation whose omission constitutes wrongful termination. Empirical studies of international EPC disputes confirm that failure to observe the cure period is among the most frequently litigated grounds in contractor termination claims (**??**).

The behavioural project management literature has studied the conditions under which organisations fail to terminate underperforming projects at the economically rational point. Escalation of commitment (**?**) — the tendency to continue investing in failing projects beyond what expected value justifies — has been documented across construction, IT, and infrastructure programmes. Real options theory (**??**) provides the normative counterpart: termination is the exercise of an abandonment option whose value exceeds continuation NPV once performance has deteriorated past a recoverable threshold. Together these streams establish that the decision to persist with or terminate an underperforming project is neither purely contractual nor purely financial — it involves performance signals, contractual obligations, and timing all at once.

The present model integrates all three dimensions into a single environment mechanic governed by two conditions. A progress delay breach fires whenever the lag between planned and actual progress exceeds the contractual tolerance, grounded in the FIDIC engineer's authority to issue a notice when progress falls materially behind programme (**?**). A hard deadline breach fires unconditionally at the contractually permitted finish overrun and triggers immediate termination regardless of the cure-period counter state. When a progress delay breach is active, the cure-period counter decrements by one; it resets to its initial value only when no breach is present. Termination executes when the counter reaches zero. This mechanic is grounded simultaneously in FIDIC contract law (**??**), the real options framing of the abandonment threshold (**??**), the EVM literature's validation of performance indices as stable forward signals (**??**), and the escalation-of-commitment literature's evidence that observable performance signals must be sustained — not transient — before termination is justified (**?**).

The cure-period counter is part of the observable state, giving the policy an anticipatory signal that reflects contractual reality without granting it discretionary control over the termination outcome. From the portfolio manager's perspective, this signal serves a dual function: it flags projects approaching contractual failure while also marking the point at which continued allocation yields diminishing returns relative to the settlement value of work already delivered. A rational manager reading a counter approaching zero in a high-opportunity-cost environment would recognise that the real value of future milestone payments is eroding while the settlement value of work in place is fixed in nominal terms — creating a natural reallocation incentive that the model captures through the interaction of the discount factor and the termination settlement mechanics, without requiring explicit inflation modelling (**??**).

2.6 Reinforcement Learning for Sequential Resource Allocation

Reinforcement learning provides a principled framework for sequential decision-making under uncertainty (**?**). The MDP formalisation — state, action, transition, reward — maps

naturally onto the portfolio budget allocation problem: the portfolio manager is modelled as a policy that observes the current state, commits an allocation, and receives a discounted net cash-flow reward, with the environment transitioning stochastically according to the realised efficiency draws.

RL has been applied to project scheduling (**??**), resource-constrained project scheduling problems (**?**), and supply chain inventory management (**?**). In each domain, RL demonstrates advantages over classical OR methods when state spaces are large, transition dynamics are stochastic, the problem is non-stationary over the decision horizon, and the horizon length is uncertain at the outset of planning. These are precisely the characteristics of the portfolio budget allocation problem.

The Gymnasium API (**?**) provides the de facto standard interface for RL environment development, enabling direct integration with modern policy optimisation libraries. Domain randomisation (**?**) — randomising environment parameters across episodes during training — is supported natively through the configuration dictionary, enabling a trained policy to generalise across diverse contract structures and project types without retraining.

Direct application of RL to portfolio-level budget allocation under contractual cash-flow mechanics remains absent from the literature. The present work provides the first MDP formulation of this problem and the first empirical demonstration that deterministic greedy heuristics systematically underperform a full-information optimum as portfolio complexity grows, establishing a quantified improvement target for future learned policies.

2.7 Synthesis: Positioning the Present Contribution

Table 1 maps representative works across the four literature streams against the key modelling dimensions of the present framework, as generated by the automated synthesis pipeline described in Section 2.1.1. Three structural gaps emerge from this mapping.

Gap 1 — Selection versus operational governance. The transition from portfolio *selection* to portfolio *operational governance* has not been modelled as a sequential decision problem. Selection literature assumes away post-selection management; project scheduling literature ignores portfolio interdependencies and the shared capital constraint. No existing work closes both simultaneously.

Gap 2 — Project-level versus portfolio-level uncertainty propagation. Existing stochastic models of project cash-flow uncertainty operate at the individual project level and do not propagate uncertainty upward into a portfolio-level decision architecture governed by a shared budget constraint. The feedback loop between a project's stochastic efficiency trajectory and its portfolio-level funding entitlement remains unmodelled.

Gap 3 — Absence of endogenous termination mechanics. The joint modelling of a progress-delay breach condition tracked against a contractual tolerance, a decrementing cure-period counter governing grace-period exhaustion, an independent hard-deadline breach condition that triggers immediate termination regardless of the cure-period state, and a termination settlement cash flow reconciling earned value against cumulative payments received, has not been embedded within a portfolio-level sequential decision framework in which budget allocation is the sole control variable.

These gaps are not isolated oversights; they reflect an incomplete progression along the path toward a deployable portfolio-level decision-support system. Figure 3 situates this progression as three sequential steps.

Step 1 — domain grounding — is well established by the literature reviewed above: contract mechanics, EVM performance signals, S-curve expenditure profiles, and

portfolio selection models collectively provide the inputs and boundary conditions that any operational governance model requires.

Step 2 — tractable problem formulation — is the gap this paper closes. The contribution is a verified, configurable MDP environment that encodes the full FIDIC cash-flow cycle and a two-condition termination mechanic, released as a Gymnasium-compatible package with a complete SQLite audit ledger. Deterministic greedy heuristics are benchmarked against a full-information optimum, producing a quantified optimality gap that grows monotonically with portfolio complexity and establishes the performance target for future learned agents.

Step 3 — validated deployment, including RL agent training and integration with incumbent project management information systems such as Oracle Primavera P6 — remains future work, contingent on the environment that Step 2 now delivers.

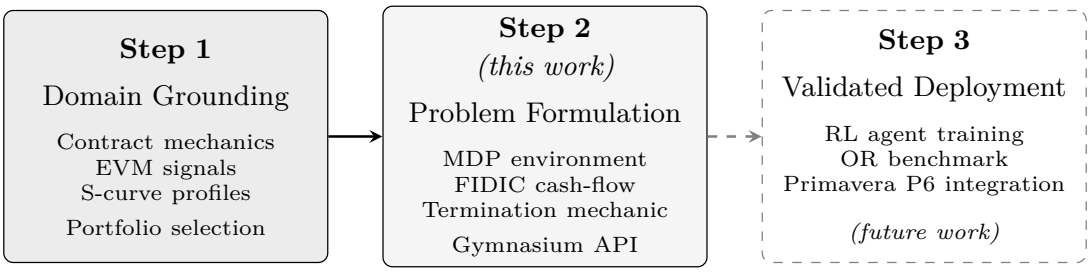


Figure 3: Maturity progression toward a deployable portfolio-level decision-support system. Step 2, completed by the present work, enables Step 3. Dashed border and arrow denote future work.

The present work addresses all three gaps and completes Step 2 of this progression. The Gymnasium-compatible MDP environment — with parameterisable contract mechanics, stochastic budget-to-progress efficiency as the single uncertainty source, and an endogenous two-condition termination mechanic — constitutes the primary methodological contribution. Deterministic greedy heuristics are evaluated against a full-information optimum in Section **??**, providing a quantified optimality gap that grows monotonically with portfolio complexity and defines the improvement target for future RL agents.

Work	Portfolio level	Decision type	Sequential decisions	Stochastic cash flows	Contract mechanics	Termination settlement
Yamakawa et al. (2019)	✓	S	–	○	–	–
Kosztyán et al. (2022)	✓	S	–	○	–	–
Touran (2006)	✓	Sc/B	–	○	–	–
Touran (2010)	✓	Sc/B	–	○	–	–
Ghapanchi et al. (2012)	✓	S	–	○	–	–
Pourahmadi et al. (2015)	✓	S	–	○	–	–
El-Sabek & McCabe (2018)	–	B	–	–	✓	–
Beseiso & Kumar (2021)	✓	S	–	○	–	–
Karimi et al. (2021)	✓	S	–	○	–	–
Bahroun et al. (2023)	–	Sc	✓	–	–	–
Ranjbar et al. (2022)	✓	S/Sc	–	–	○	–
Szwarcfiter et al. (2023)	✓	Sc	✓	○	–	–
Asghari et al. (2023)	–	–	✓	○	–	–
Mazhari & Tajik (2024)	✓	S/B	✓	–	–	–
Jiang et al. (2024)	✓	–	✓	○	–	–
Nabot et al. (2025)	✓	Sc	✓	–	✓	✓
Darvish & Sepehri (2025)	✓	B	✓	○	○	✓
Gholami et al. (2025)	✓	Sc	✓	○	–	✓
Kehinde (2025)	✓	B	✓	✓	–	–
Ajibade (2025)	✓	Sc	✓	○	–	–
Huddar & Joshi (2025)	✓	B	✓	○	–	–
Akbari & Akbarpour Shirazi (2026)	✓	B	✓	○	✓	✓
Ebnerasoul et al. (2023)	✓	S	–	○	–	–
Noroozi et al. (2016)	✓	B	–	○	–	–
Jiang et al. (2023)	–	B	✓	–	–	–
Present work	✓	**B**	✓	✓	✓	✓

Table 1: Representative literature mapped against modelling dimensions of the present framework. *Decision type*: S = selection, Sc = scheduling, B = budgeting. ✓ = fully addressed; ○ = partially addressed; – = not addressed.

3 Problem Formulation

This section develops the complete mathematical specification of the portfolio budget allocation problem and its realisation as a Gymnasium-compatible reinforcement learning environment. Section **??** states the scope and structural assumptions. Section **??** defines the environment primitives: progress dynamics, FIDIC cash-flow mechanics, EVM performance signals, the termination mechanic, and the treasury balance. Section **??** consolidates all parameters. Section **??** derives the full-foresight upper bound and establishes why the exact stochastic problem is computationally intractable at portfolio scale. Section **??** presents the MDP formulation: state space, action space, action projection, transition kernel, reward function, and the observation vector that constitutes the environment's interface to any policy.

3.1 Problem Statement and Scope

The problem is at strategic governance level. A portfolio of N contracted projects is fixed: commitments are signed, start dates are scheduled, and contracts are parameterised consistently with FIDIC Red, Yellow, and Silver Book forms (**?**). A portfolio manager exercises a single recurring decision at each discrete planning period $t \in \{0, 1, \ldots, T\}$: the monetary budget $x_{i,t} \geq 0$ directed to each active project i. The episode horizon is

$$T = \max_i \big(f_i + \Delta_i^{\max}\big) - 1, \tag{1}$$

where $f_i = s_i + D_i$ is the planned finish and $\Delta_i^{\max}$ is the maximum permitted finish overrun for project i; period $t = 0$ aligns with the earliest planned project start. All outcomes of interest — milestone achievement, cash inflows, schedule adherence, and project termination — flow through the allocation decision. Portfolio selection is treated as an exogenous input (**?**).

The formulation rests on four structural assumptions.

(i) *Independent bilateral contracts.* Projects interact only through the shared portfolio cash balance; no resource or workforce sharing occurs, consistent with standard EPC contracting practice.

(ii) *Budget as the sole control.* The contractor's internal resource decisions are absorbed into a stochastic efficiency scalar $\eta_{i,t} \geq 0.01$; the manager observes how much physical progress a budget allocation yields, not how it is achieved. $\eta_{i,t}$ is the sole stochastic primitive and is not observable before $x_{i,t}$ is committed.

(iii) *Discrete monthly periods.* The unit period is one calendar month, reflecting the standard governance cycle. A budget tightness ratio $\kappa = B_0 / \sum_i \mathrm{BAC}_i$ parameterises the liquidity regime.

(iv) *FIDIC contract mechanics.* Advance payment and recovery, milestone revenues, retention withholding and release, and termination settlement are all configurable, accommodating the full range of particular conditions encountered in practice.

3.2 Environment Primitives

3.2.1 Progress and Expenditure Dynamics

Let τ_i denote the project-time elapsed for project i since its planned start s_i, incremented by one each period the project is active. Budget $x_{i,t}$ converts to physical progress through

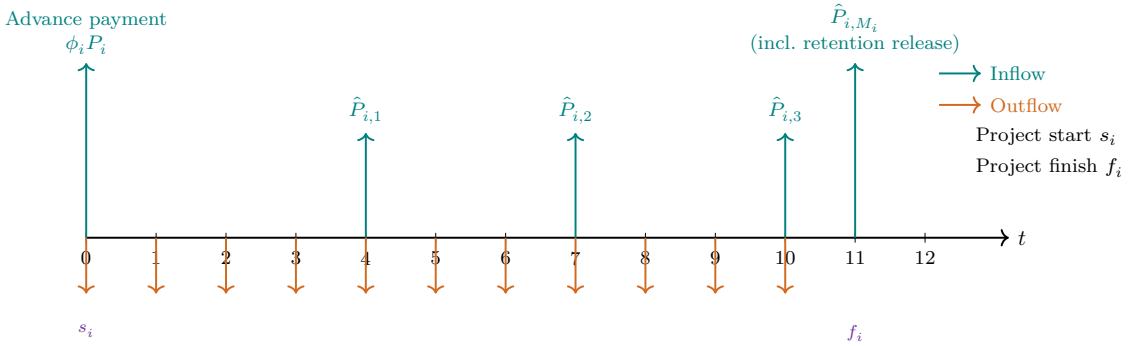


Figure 4: Cash-flow timeline for project i under FIDIC contract mechanics. The budget-to-progress conversion is subject to stochastic efficiency $\eta_{i,t}$, the sole source of uncertainty in the model. Net milestone payments $\hat{P}_{i,j}$ (**??**) are received at certification; the final payment includes the full retention release.

the stochastic efficiency scalar $\eta_{i,t}$:

$$EV_{i,t} = \min\Big(1,\ EV_{i,t-1} + \eta_{i,t} \cdot \frac{x_{i,t}}{\mathrm{BAC}_i}\Big), \quad \mathrm{ACWP}_{i,t} = \mathrm{ACWP}_{i,t-1} + x_{i,t}, \tag{2}$$

with $EV_{i,0} = \mathrm{ACWP}_{i,0} = 0$. Here BAC_i is the Budget at Completion and $EV_{i,t} \in [0,1]$ is the fraction of work completed (Budgeted Cost of Work Performed divided by BAC_i). The progress increment $\eta_{i,t} x_{i,t}/\mathrm{BAC}_i$ is realised only after allocation is committed; $\eta_{i,t}$ is drawn from a project-specific distribution whose family and parameters are supplied in the configuration dictionary.

The *planned* cumulative progress follows a normalised S-curve parameterised by Beta shape parameters (a_i, b_i) (**?**):

$$PV_i(\tau) = I_{\tau/D_i}(a_i,\, b_i), \quad \tau \in [0, D_i], \quad D_i \geq 1, \tag{3}$$

where $I_u(\cdot,\cdot)$ is the regularised incomplete Beta function and $D_i \in \mathbb{Z}_{\geq 1}$ is the planned duration in periods. The minimum admissible duration is one period ($D_i = 1$), which models a project whose entire planned scope must be delivered within a single allocation at $t = s_i$: the Beta CDF maps $\tau = 0 \mapsto 0$ and $\tau = 1 \mapsto 1$, so the manager has exactly one period to commit the full allocation before the hard deadline fires at $t = s_i + 1$. $PV_i(\tau_i)$ is the planned progress at the current project-time; $PV_i(\tau_i + 1)$ and $PV_i(\tau_i + 2)$ serve as one- and two-period lookaheads used in the observation vector and the baseline policies.

3.2.2 FIDIC Cash-Flow Mechanics

Contract price and advance payment. The contract price is $P_i = \mathrm{BAC}_i(1 + \pi_i)$, where π_i is the profit margin. If the contract includes an advance payment ($\phi_i > 0$), the portfolio receives at project start ($t = s_i$):

$$\mathrm{Adv}_i = \phi_i \cdot P_i, \tag{4}$$

where ϕ_i is the advance payment rate. When $\phi_i = 0$ no advance is disbursed, the recovery mechanism is inactive, and $A^{\mathrm{rec}}_{i,j} = 0$ for all milestones. The advance is modelled as a synthetic milestone $j = 0$ with progress threshold zero, certified automatically at $t = s_i$, so that $\hat{P}_{i,0} = \mathrm{Adv}_i$ enters V_{s_i} through (**??**) without special-casing.

Recovery of the advance from subsequent milestone payments follows the FIDIC Sub-Clause 14.2 mechanism (**?**), governed by two project-specific parameters: a commencement threshold $\psi_i^{\text{rec}} \in [0,1]$ and a per-payment deduction rate $\delta_i^{\text{rec}} \in [0,1]$. Let $G_{i,j}^{\text{cum}} = \sum_{k=1}^{j} G_{i,k}$ denote the cumulative gross certified payment through milestone j. The advance recovery deduction at milestone j is:

$$A_{i,j}^{\text{rec}} = \begin{cases} \min\left(\delta_i^{\text{rec}} \cdot G_{i,j},\ \text{Adv}_{i,j}^{\text{rem}}\right) & \text{if } G_{i,j-1}^{\text{cum}} \geq \psi_i^{\text{rec}} \cdot P_i \text{ and } \phi_i > 0, \\ 0 & \text{otherwise,} \end{cases} \tag{5}$$

where $\text{Adv}_{i,j}^{\text{rem}}$ is the advance balance outstanding before milestone j is certified, ensuring the total recovered never exceeds Adv_i. When $\psi_i^{\text{rec}} = 0$, recovery deductions commence from the very first milestone payment.

Milestone certification and net payments. Milestones $j = 1, \ldots, M_i$ are ordered by progress threshold $\psi_{i,1} < \cdots < \psi_{i,M_i} = 1$. Milestone j is certified at the first period t satisfying both $EV_{i,t} \geq \psi_{i,j}$ and $t \geq s_i + \tau_{i,j}^{\text{ms}}$, where $\tau_{i,j}^{\text{ms}}$ is the project-time threshold enforcing the FIDIC Sub-Clause 14.3 payment interval (**?**). Certification is monotone. The net cash received upon certifying milestone j is:

$$\hat{P}_{i,j} = G_{i,j} - A_{i,j}^{\text{rec}} - R_{i,j}^{\text{held}} + R_{i,j}^{\text{rel}}, \tag{6}$$

where $G_{i,j} = w_{i,j} P_i$ is the gross payment ($w_{i,j}$ is the milestone weight, $\sum_{j=1}^{M_i} w_{i,j} = 1$); $A_{i,j}^{\text{rec}}$ is the advance recovery deduction (**??**); $R_{i,j}^{\text{held}} = \nu_i G_{i,j}$ is the retention withheld; and $R_{i,j}^{\text{rel}}$ equals the total accumulated retention only at the final milestone, zero otherwise. When $\phi_i > 0$, the full payment identity $\text{Adv}_i + \sum_{j=1}^{M_i} \hat{P}_{i,j} = P_i$ holds exactly; when $\phi_i = 0$ it reduces to $\sum_{j=1}^{M_i} \hat{P}_{i,j} = P_i$. Payments are received in the period of certification.

Period aggregate inflow. The total portfolio inflow at period t is:

$$V_t = \sum_{i=0}^{N-1} \sum_{j=0}^{M_i} \hat{P}_{i,j} \cdot \mathbf{1}\left[t_{i,j}^{\text{cert}} = t\right] + \sum_{i \in \mathcal{P}_t^{\text{term}}} S_i, \tag{7}$$

where $t_{i,j}^{\text{cert}}$ is the certification period of milestone j (null if uncertified), $\mathcal{P}_t^{\text{term}}$ collects projects newly terminated at t, and S_i is the termination settlement defined in Section **??**.

3.2.3 EVM Performance Signals

The Earned Value Management framework (**?**) provides four diagnostic indices computed each period from observable state. The Earned Schedule $\text{ES}_{i,t}$ is obtained by inverting the planned S-curve at the current actual progress level: it is the project-time $\tau^* \in [0, D_i]$ such that $I_{\tau^*/D_i}(a_i, b_i) = EV_{i,t}$, computed by binary search. The four indices are:

$$\text{SPI}_{i,t} = \frac{\text{ES}_{i,t}}{\tau_i}, \tag{8}$$

$$\text{CPI}_{i,t} = \frac{EV_{i,t} \cdot \text{BAC}_i}{\text{ACWP}_{i,t}}, \tag{9}$$

$$\text{EAC}_{i,t} = \text{ACWP}_{i,t} + \frac{\text{BAC}_i - EV_{i,t} \cdot \text{BAC}_i}{\text{CPI}_{i,t} \cdot \text{SPI}_{i,t}}, \tag{10}$$

$$\text{TCPI}_{i,t} = \frac{\text{BAC}_i - EV_{i,t} \cdot \text{BAC}_i}{\text{BAC}_i - \text{ACWP}_{i,t}}. \tag{11}$$